Storyline

Chapter 7: Energy of a System



Physics for Scientists and Engineers, 10e
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BROOKS/COLE
CENGAGE Learning

Systems and Environments

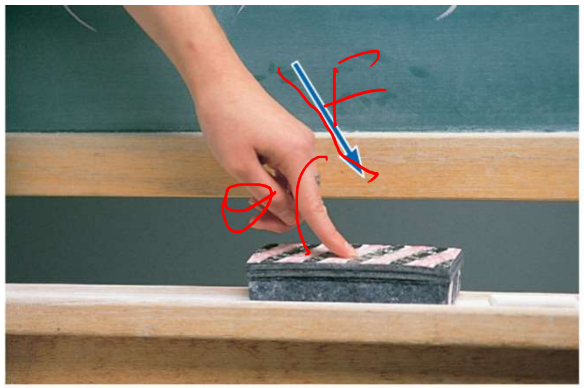
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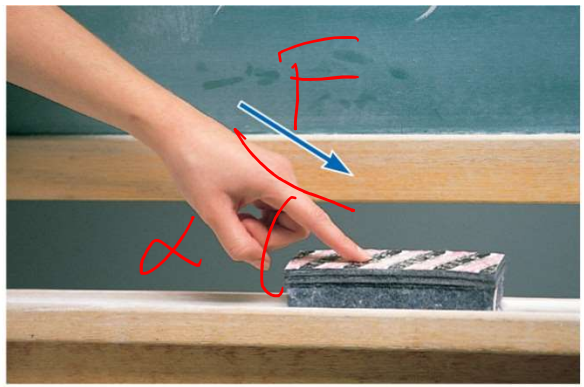


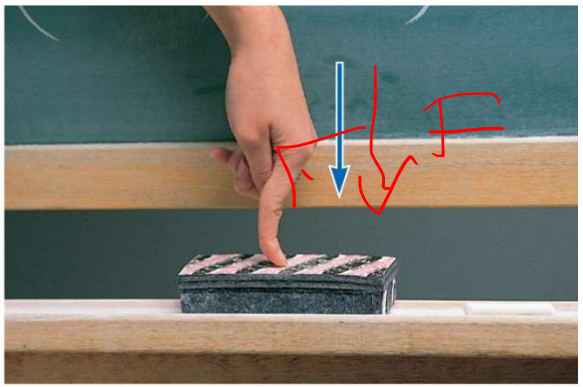
- may be a single object or particle
- may be a collection of objects or particles
- may be a region of space (such as the interior of an automobile engine combustion cylinder)
- may vary with time in size and shape (such as a rubber ball, which deforms upon striking a wall)

Work Done by a Constant Force

F is constant not changing with time

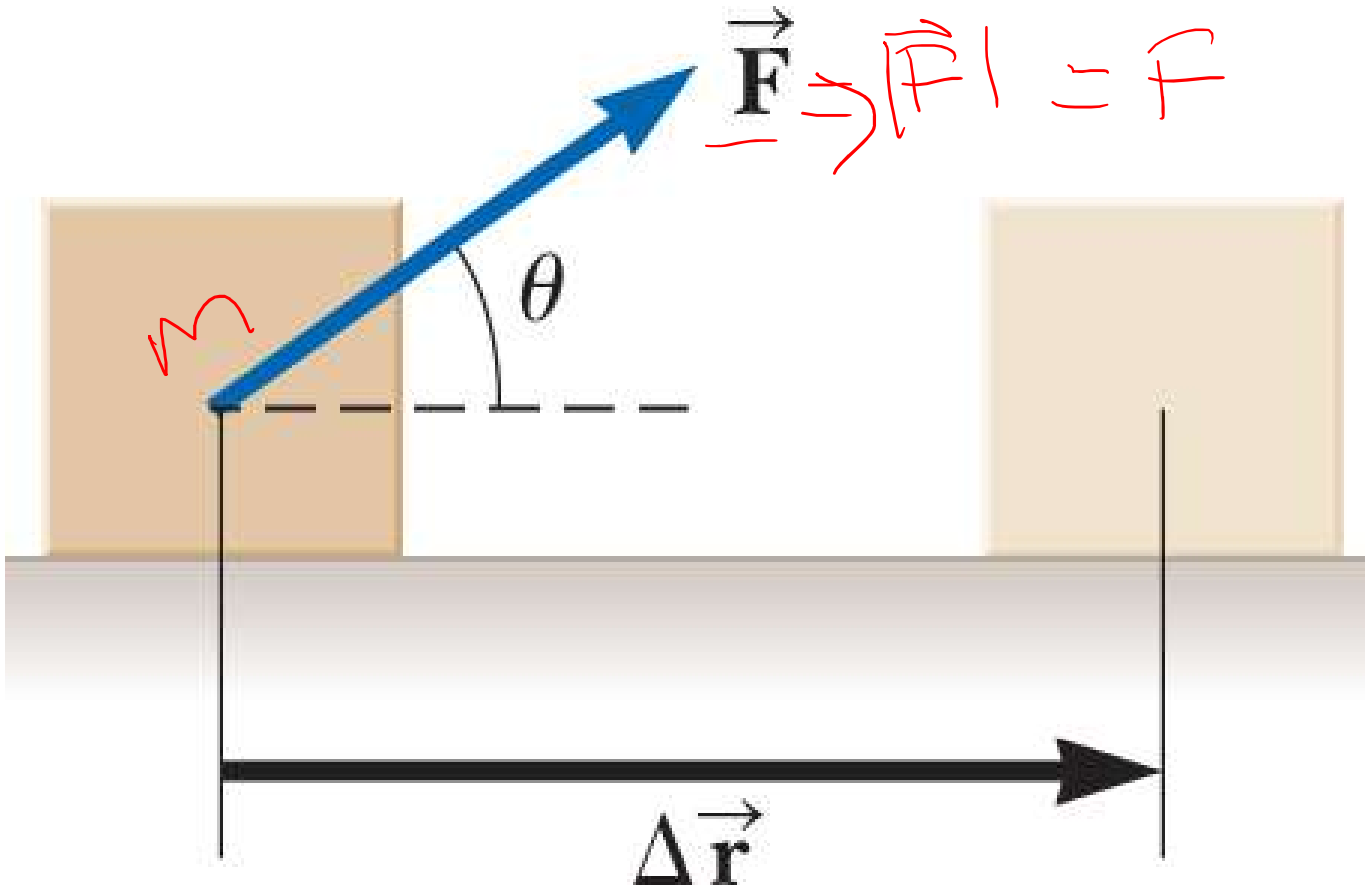




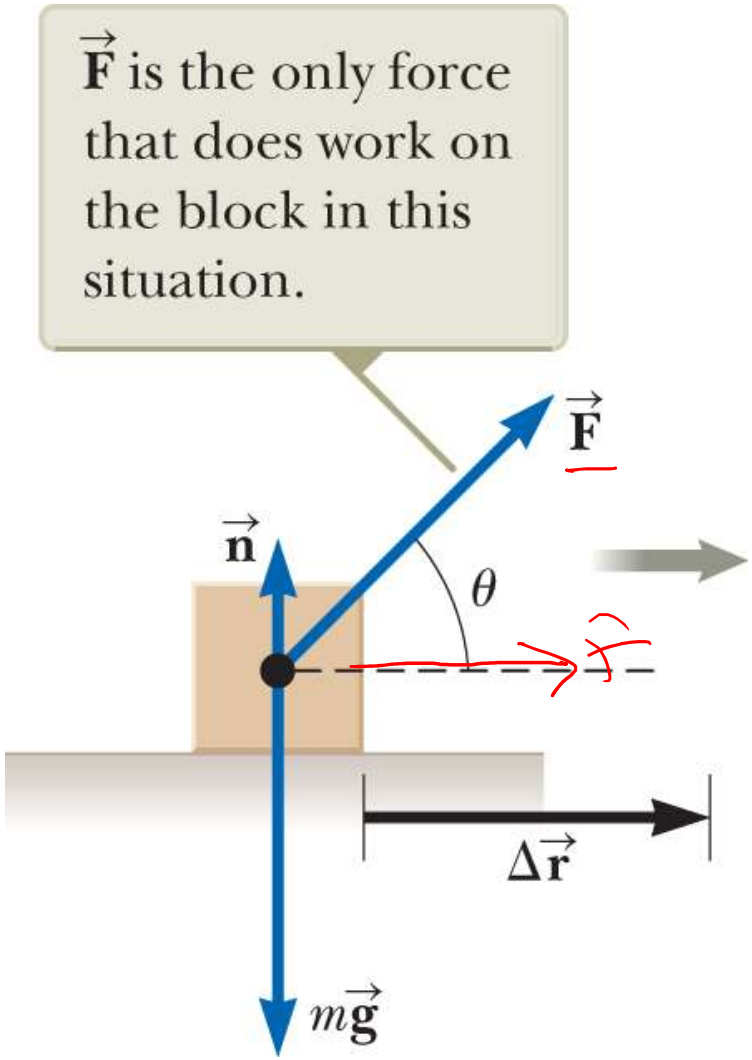


Work Done by a Constant Force

$$W \equiv F\Delta r \cos\theta$$



Work Done by a Constant Force



$$W \equiv F\,\Delta r\cos\theta$$

joule = N·m

$$\text{for } \theta = 0 \Rightarrow W = F\,\Delta r$$

$$\mathrm{N\cdot m} \equiv \mathrm{J\ (joule)} = \mathrm{kg\cdot m^2/s^2}$$

Work: Energy Transfer

Work is an energy transfer



Quick Quiz 7.1

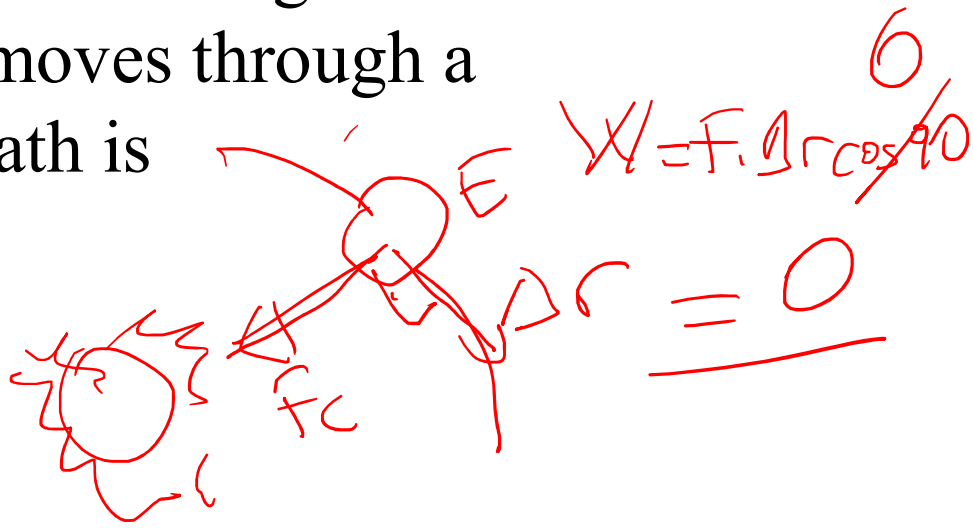
The gravitational force exerted by the Sun on the Earth holds the Earth in an orbit around the Sun. Let us assume that the orbit is perfectly circular. The work done by this gravitational force during a short time interval in which the Earth moves through a displacement in its orbital path is

(a) zero

(b) positive

(c) negative

(d) impossible to determine



Quick Quiz 7.1

The gravitational force exerted by the Sun on the Earth holds the Earth in an orbit around the Sun. Let us assume that the orbit is perfectly circular. The work done by this gravitational force during a short time interval in which the Earth moves through a displacement in its orbital path is

(a) zero

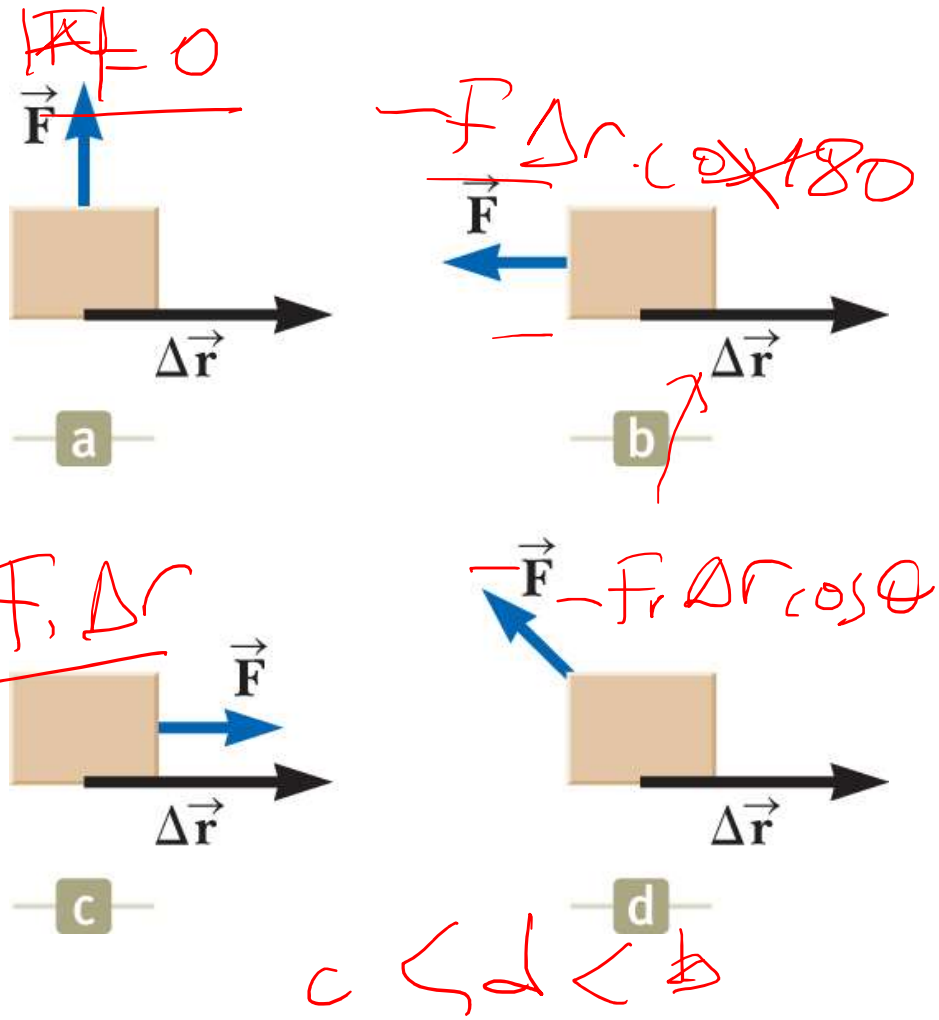
(b) positive

(c) negative

(d) impossible to determine

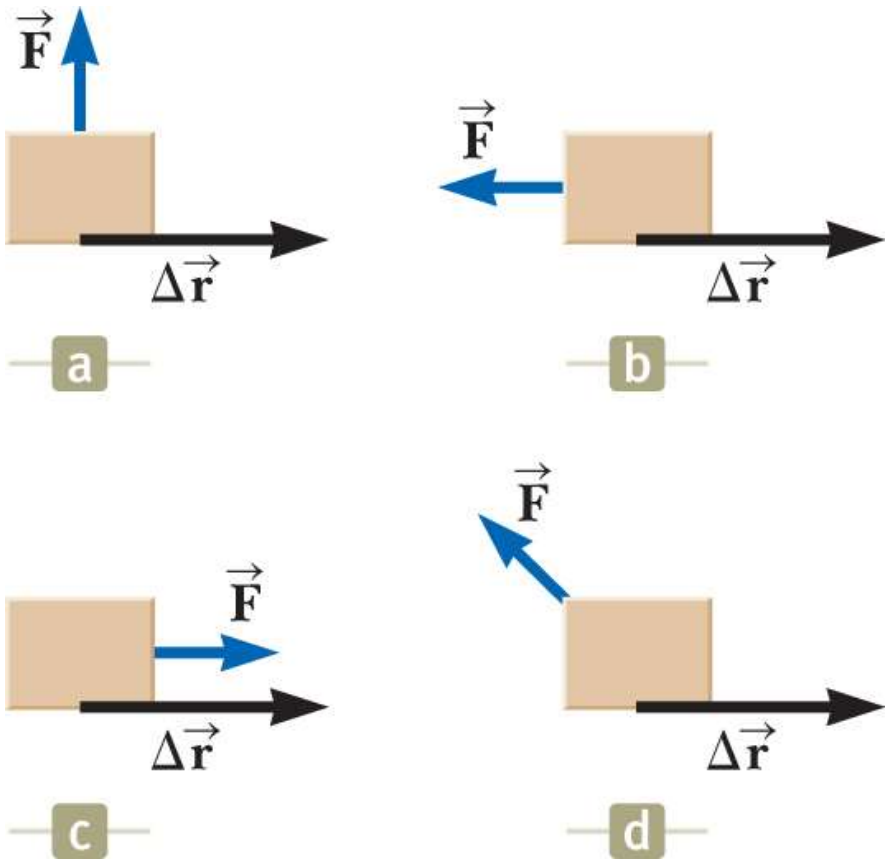
Quick Quiz 7.2

The figure shows four situations in which a force is applied to an object. In all four cases, the force has the same magnitude, and the displacement of the object is to the right and of the same magnitude. Rank the situations in order of the work done by the force on the object, from most positive to most negative.



Quick Quiz 7.2

The figure shows four situations in which a force is applied to an object. In all four cases, the force has the same magnitude, and the displacement of the object is to the right and of the same magnitude. Rank the situations in order of the work done by the force on the object, from most positive to most negative.



(c), (a), (d), (b)

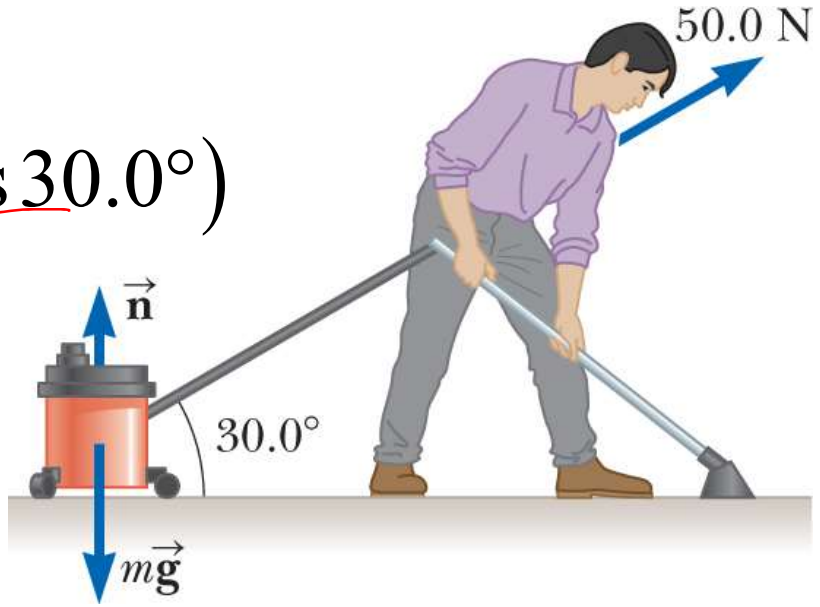
Example 7.1:
Mr. Clean

A man cleaning a floor pulls a vacuum cleaner with a force of magnitude $F = 50.0$ N at an angle of 30.0° with the horizontal. Calculate the work done by the force on the vacuum cleaner as the vacuum cleaner is displaced 3.00 m to the right.

$$W = F\Delta r\cos\theta$$

$$= (50.0\text{ N})(3.00\text{ m})(\cos 30.0°)$$

$$= \boxed{130\text{ J}}$$



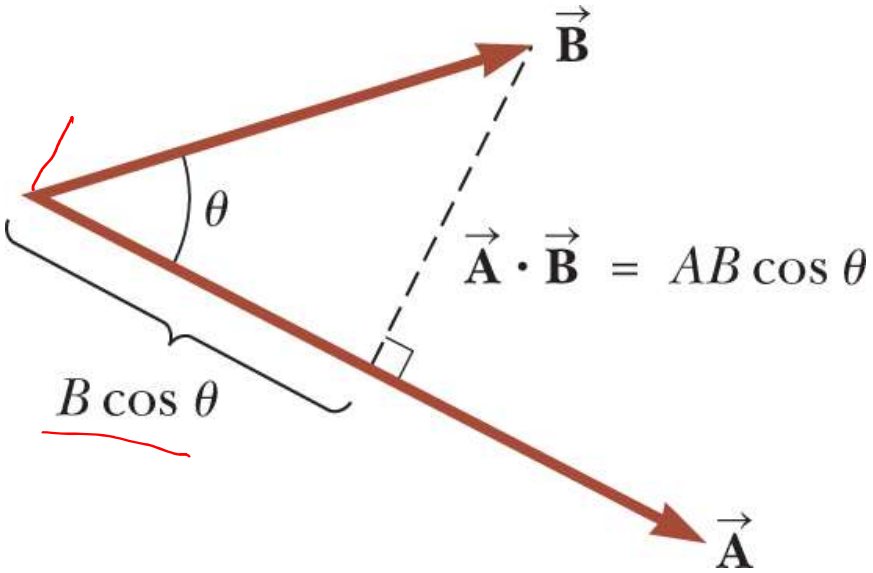
The Scalar Product of Two Vectors

$$W = \vec{F} \cdot \Delta\vec{r}$$

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} \equiv AB\cos\theta$$

$$W = F\Delta r\cos\theta = \vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{r}}$$

The Scalar Product of Two Vectors



$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} \equiv AB\cos\theta$$

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = \vec{\mathbf{B}} \cdot \vec{\mathbf{A}}$$

$$\vec{\mathbf{A}} \cdot \left(\vec{\mathbf{B}} + \vec{\mathbf{C}}\right) = \vec{\mathbf{A}} \cdot \vec{\mathbf{B}} + \vec{\mathbf{A}} \cdot \vec{\mathbf{C}}$$

$$\vec{\mathbf{A}} \perp \vec{\mathbf{B}} \quad (\theta = 90°) \Rightarrow \vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = 0$$

$$\vec{\mathbf{A}} \parallel \vec{\mathbf{B}} \quad (\theta = 0°) \Rightarrow \vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = AB$$

$$\vec{\mathbf{A}} \parallel \vec{\mathbf{B}} \quad (\theta = 180°) \Rightarrow \vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = -AB$$

The Scalar Product of Two Vectors

$$\hat{\mathbf{i}}\cdot\hat{\mathbf{i}} = \hat{\mathbf{j}}\cdot\hat{\mathbf{j}} = \hat{\mathbf{k}}\cdot\hat{\mathbf{k}} = 1$$

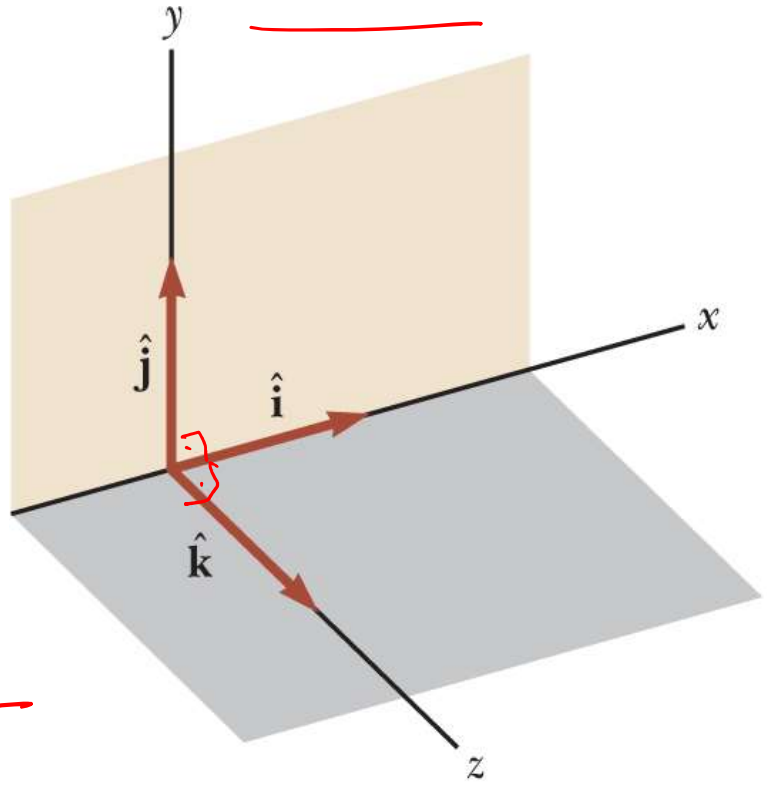
$$\hat{\mathbf{i}}\cdot\hat{\mathbf{j}} = \hat{\mathbf{i}}\cdot\hat{\mathbf{j}} = \hat{\mathbf{j}}\cdot\hat{\mathbf{k}} = 0$$

$$\vec{\mathbf{A}} = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}$$

$$\vec{\mathbf{B}} = B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}}$$

$$\vec{\mathbf{A}}\cdot\vec{\mathbf{B}} = A_xB_x + A_yB_y + A_zB_z$$

$$\vec{\mathbf{A}}\cdot\vec{\mathbf{A}} = A_x^2 + A_y^2 + A_z^2 = A^2$$



Quick Quiz 7.3

Which of the following statements is true about the relationship between the dot product of two vectors and the product of the magnitudes of the vectors?

(a) $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ is larger than AB.

(b) $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ is smaller than AB.

(c) $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ could be larger or smaller than AB, depending on the angle between the vectors.

(d) $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ could be equal to AB.

Quick Quiz 7.3

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(a) $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ is larger than AB.

(b) $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ is smaller than AB.

(c) $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ could be larger or smaller than AB, depending on the angle between the vectors.

(d) **$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$ could be equal to AB.**

Example 7.2:
The Scalar Product

The vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ are given by

$$\vec{\mathbf{A}} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} \text{ and } \vec{\mathbf{B}} = -\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$$

(A) Determine the scalar product $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$.

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = \left(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}\right) \cdot \left(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}}\right)$$

$$= -2\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} + 2\hat{\mathbf{i}} \cdot 2\hat{\mathbf{j}} - 3\hat{\mathbf{j}} \cdot \hat{\mathbf{i}} + 3\hat{\mathbf{j}} \cdot 2\hat{\mathbf{j}}$$

$$= 2(1) + 4(0) - 3(0) + 6(1) = -2 + 6 = \boxed{4}$$

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = A_x B_x + A_y B_y = (2)(-1) + (3)(2) = 4$$

Example 7.2:
The Scalar Product

(B) Find the angle θ between $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$.

$$A = \sqrt{A_x^{\,2} + A_y^{\,2}} = \sqrt{(2)^2 + (3)^2} = \sqrt{13}$$

$$\mathbf{B} = \sqrt{B_x^{\,2} + B_y^{\,2}} = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$$

$$\cos\theta = \frac{\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}}{AB} = \frac{4}{\sqrt{13}\sqrt{5}} = \frac{4}{\sqrt{65}}$$

$$\theta = \cos^{-1}\frac{4}{\sqrt{65}} = 60.3°$$

Example 7.3:
Work Done by a Constant Force

A particle moving in the *xy* plane undergoes a displacement given by $\vec{\mathbf{r}} = \left(2.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}}\right)$ m as a constant force $\vec{\mathbf{F}} = \left(5.0\hat{\mathbf{i}} + 2.0\hat{\mathbf{j}}\right)$ N acts on the particle. Calculate the work done by $\vec{\mathbf{F}}$ on the particle.

$$W = \vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{r}} = \left[\left(5.0\hat{\mathbf{i}} + 2.0\hat{\mathbf{j}}\right)\ \text{N}\right] \cdot \left[\left(2.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}}\right)\ \text{m}\right]$$

$$= \left(5.0\hat{\mathbf{i}} \cdot 2.0\hat{\mathbf{i}} + 5.0\hat{\mathbf{i}} \cdot 3.0\hat{\mathbf{j}} + 2.0\hat{\mathbf{j}} \cdot 2.0\hat{\mathbf{i}} + 2.0\hat{\mathbf{j}} \cdot 3.0\hat{\mathbf{j}}\right)\ \text{N} \cdot \text{m}$$

$$= \left[10 + 0 + 0 + 6\right]\ \text{N} \cdot \text{m} = \boxed{16\ \text{J}}$$

Work Done by a Varying Force

$$W \approx F_x \Delta x$$

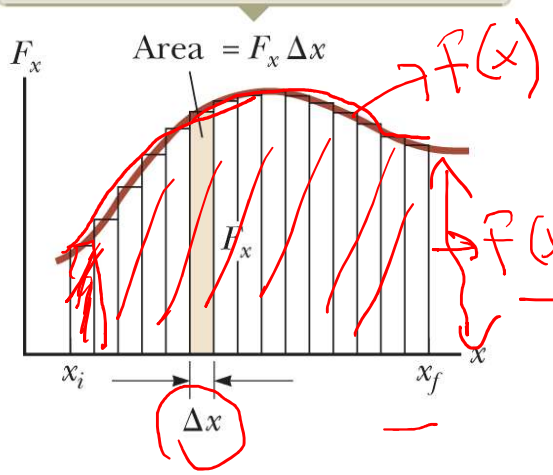
$$W \approx \sum_{x_i}^{x_f} F_x \Delta x$$

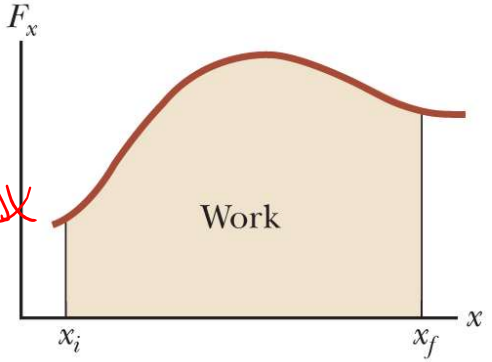
$$\lim_{\Delta x \to 0} \sum_{x_i}^{x_f} F_x \Delta x = \int_{x_i}^{x_f} F_x \, dx$$

$$W = \int_{x_i}^{x_f} F_x \, dx$$

The total work done for the displacement from x_i to x_f is approximately equal to the sum of the areas of all the rectangles.

The work done by the component F_x of the varying force as the particle moves from x_i to x_f is *exactly* equal to the area under the curve.





Work Done by a Varying Force

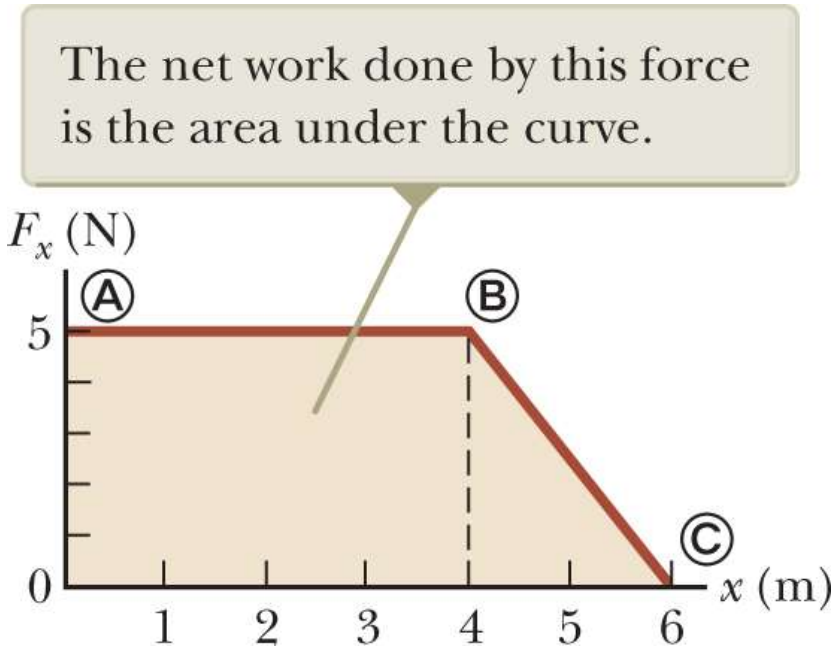
$$\sum W = W_{\text{ext}} = \int_{x_i}^{x_f} \left(\sum F_x \right) dx \quad \text{(particle)}$$

$$\sum W = W_{\text{ext}} = \int \left(\sum F_x \right) \cdot d\vec{\mathbf{r}} \quad \text{(particle)}$$

$$\sum W = W_{\text{ext}} = \sum_{\text{forces}} \left(\int \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \right) \quad \text{(deformable system)}$$

Example 7.4:
Calculating Total Work Done from a Graph

A force acting on a particle varies with x as shown in the figure. Calculate the work done by the force on the particle as it moves from $x = 0$ to $x = 6.0$ m.



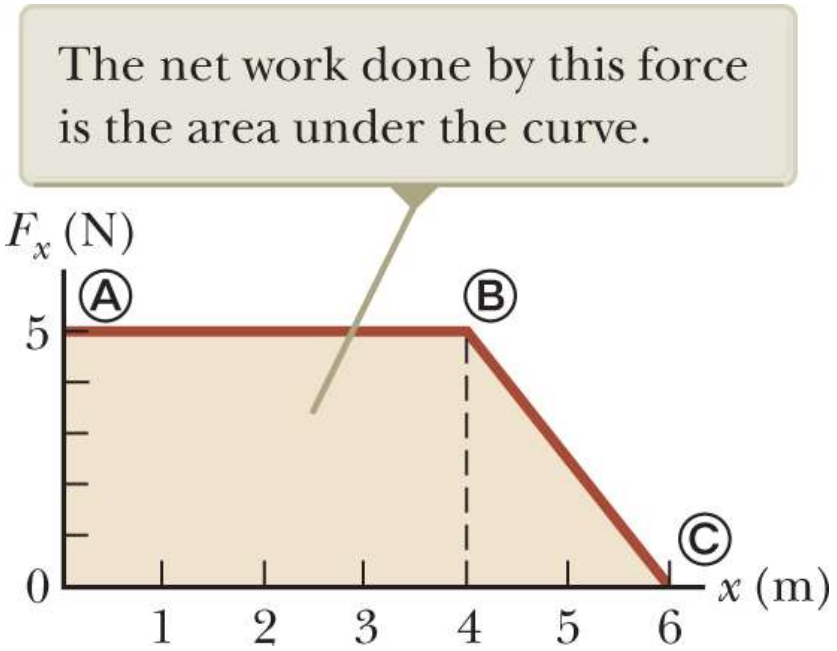
Example 7.4:
Calculating Total Work Done from a Graph

$$W_{\text{A to B}} = (5.0\ \text{N})(4.0\ \text{m}) = 20\ \text{J}$$

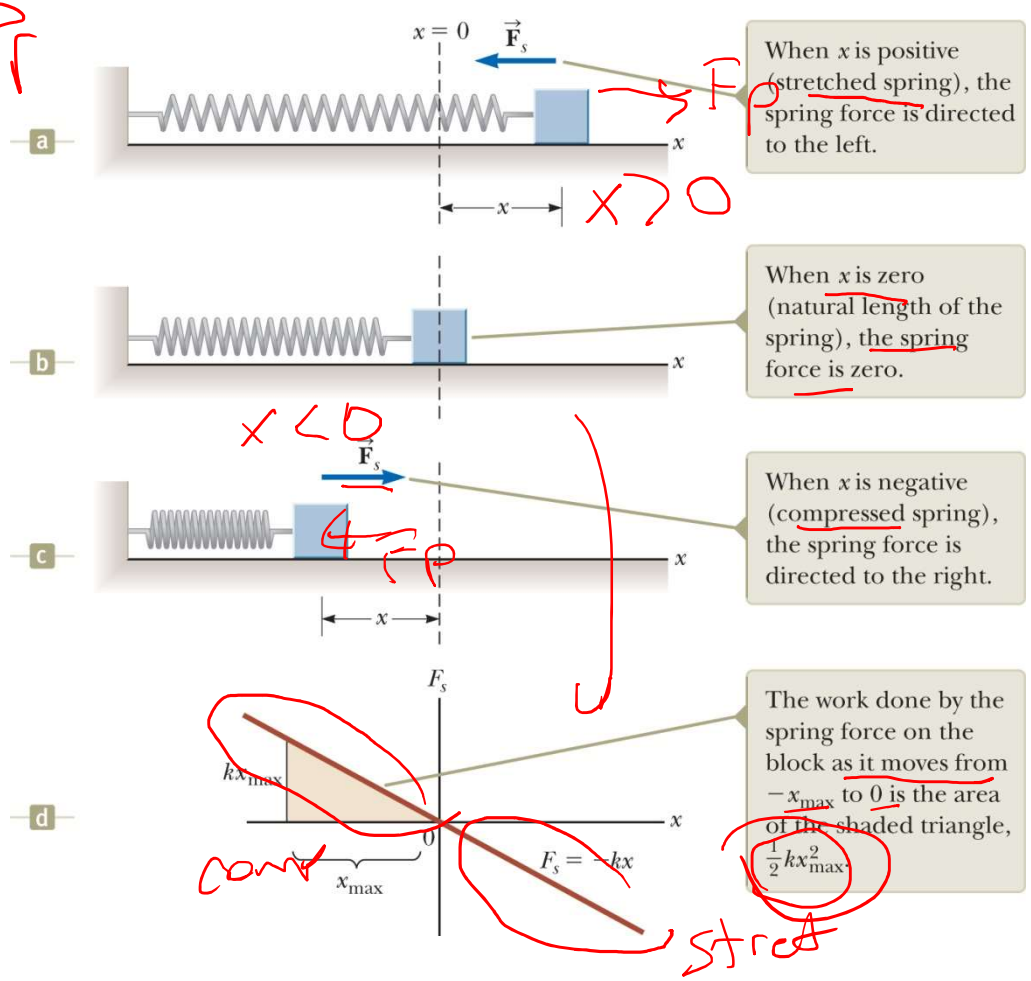
$$W_{\text{B to C}} = \frac{1}{2}(5.0\ \text{N})(2.0\ \text{m}) = 5.0\ \text{J}$$

$$W_{\text{A to C}} = W_{\text{A to B}} + W_{\text{B to C}}$$

$$= 20\ \text{J} + 5.0\ \text{J} = \boxed{25\ \text{J}}$$



Work Done by a Spring



$$F_s = -kx$$

$$\vec{\mathbf{F}}_s = F_s\hat{\mathbf{i}} = -kx\hat{\mathbf{i}}$$

Work Done by a Spring

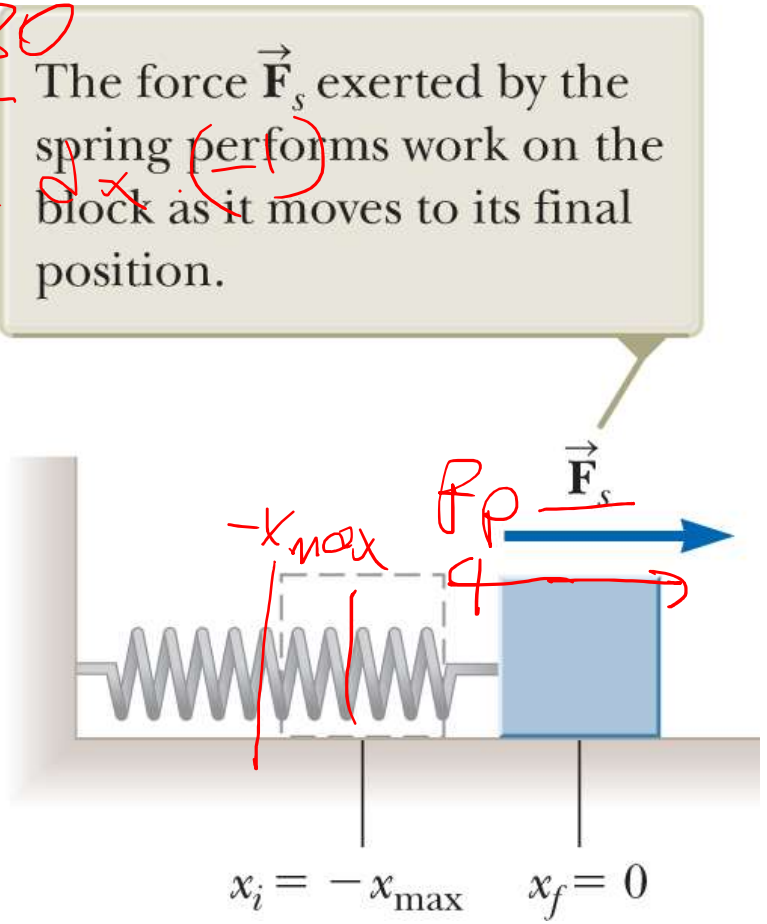
$$W_s = \int \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

$$= \int_{x_i}^{x_f} \left(-kx\hat{\mathbf{i}}\right) \cdot \left(dx\hat{\mathbf{i}}\right)$$

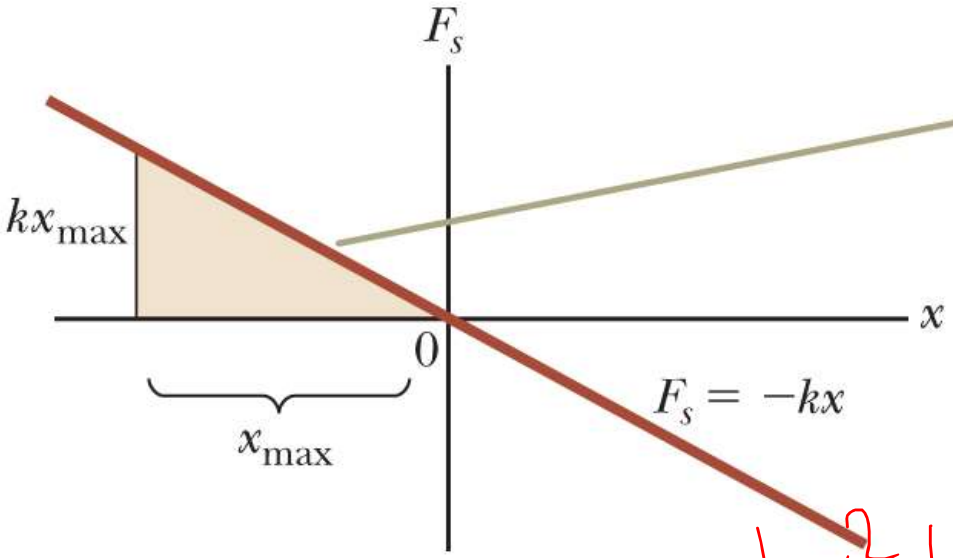
$$= \int_{-x_{\max}}^{0} \left(-kx\right) dx = \frac{1}{2} kx_{\max}^2$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$W_s = \frac{1}{2} kx_{\max}^2$$



Work Done by a Spring



The work done by the spring force on the block as it moves from $-x_{max}$ to 0 is the area of the shaded triangle, $\frac{1}{2}kx_{max}^2$.

$$W_s = \int_{x_i}^{x_f} (-kx)\,dx = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$$

Work Done by a Spring

$$F_{\text{app}} = F_{\text{app}}\hat{\mathbf{i}} = -\vec{\mathbf{F}}_s$$

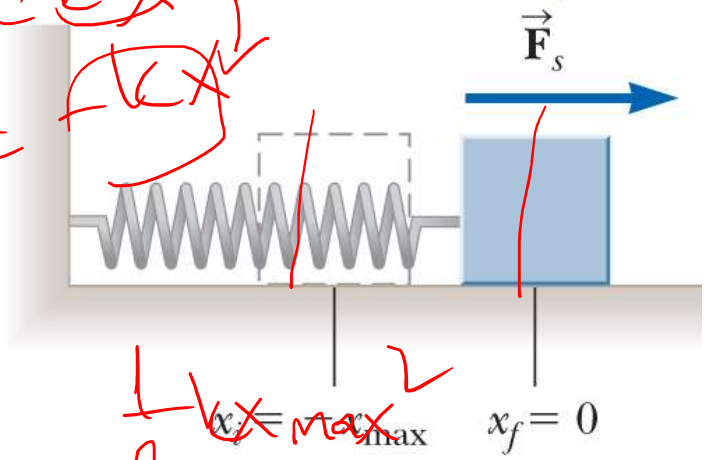
$$= -\left(-kx\hat{\mathbf{i}}\right) = kx\hat{\mathbf{i}}$$

$$W_{\text{ext}} = \int \vec{\mathbf{F}}_{\text{app}} \cdot d\vec{\mathbf{r}}$$

$$= \int_{x_i}^{x_f} \left(kx\hat{\mathbf{i}}\right) \cdot \left(dx\hat{\mathbf{i}}\right)$$

$$= \int_{-x_{\max}}^{0} kx\,dx = -\frac{1}{2}kx_{\max}^2$$

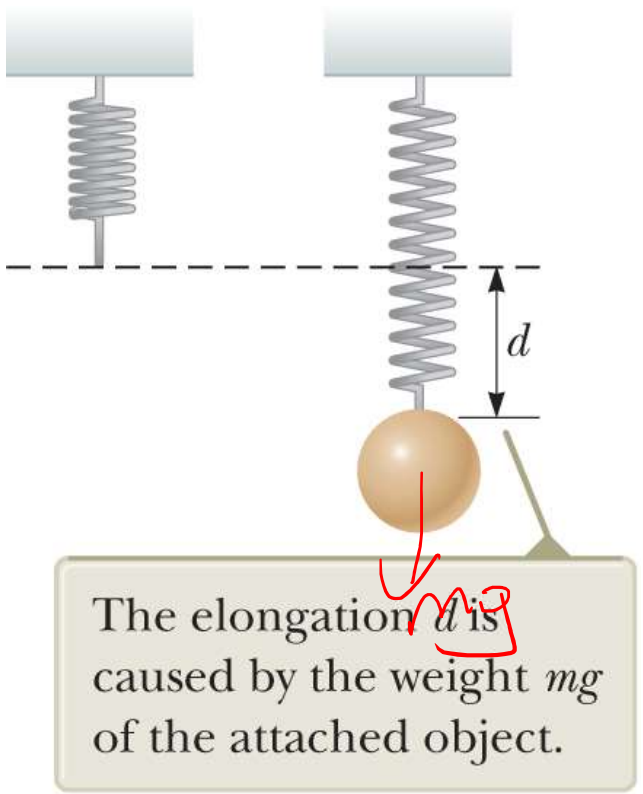
The force $\vec{\mathbf{F}}_s$ exerted by the spring performs work on the block as it moves to its final position.



$$W_{\text{ext}} = \int_{x_i}^{x_f} kx\,dx = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2$$

Example 7.5: Measuring *k* for a Spring

A common technique used to measure the force constant of a spring is demonstrated by the setup in the figure. The spring is hung vertically, and an object of mass *m* is attached to its lower end. Under the action of the "load" *mg*, the spring stretches a distance *d* from its equilibrium position.



Example 7.5: Measuring *k* for a Spring

(A) If a spring is stretched 2.0 cm by a suspended object having a mass of 0.55 kg, what is the force constant of the spring?

$F_{net} = 0 = ma = 0$

$$\vec{\mathbf{F}}_x + m\vec{\mathbf{g}} = 0 \rightarrow F_s - mg = 0$$

$$\rightarrow F_s = mg \qquad F_S = mg = kx$$

$$k = \frac{mg}{d} = \frac{(0.55\ \text{kg})(9.80\ \text{m/s}^2)}{2.0\times10^{-2}\ \text{m}}$$

$$= \boxed{2.7\times10^{2}\ \text{N/m}}$$ ✓



Example 7.5: Measuring *k* for a Spring

(B) How much work is done by the spring on the object as it stretches through this distance?

$$W_s = 0 - \frac{1}{2}kd^2$$

$$= -\frac{1}{2}\left(2.7\times10^{2}\ \text{N/m}\right)\left(2.0\times10^{-2}\ \text{m}\right)^2$$

$$= -5.4\times10^{-2}\ \text{J}$$

Example 7.5: Measuring *k* for a Spring

$$W = \vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{r}} = (mg)(d)\cos 0 = mgd$$

$$= (0.55\ \text{kg})(9.80\ \text{m/s}^2)(2.0\times10^{-2}\ \text{m}) = 1.1\times10^{-1}\ \text{J}$$

$F_{ext} = mg$

mg

$\theta = 0$

$\cos 0$

Work done by mg

Kinetic Energy and the Work–Kinetic Energy Theorem

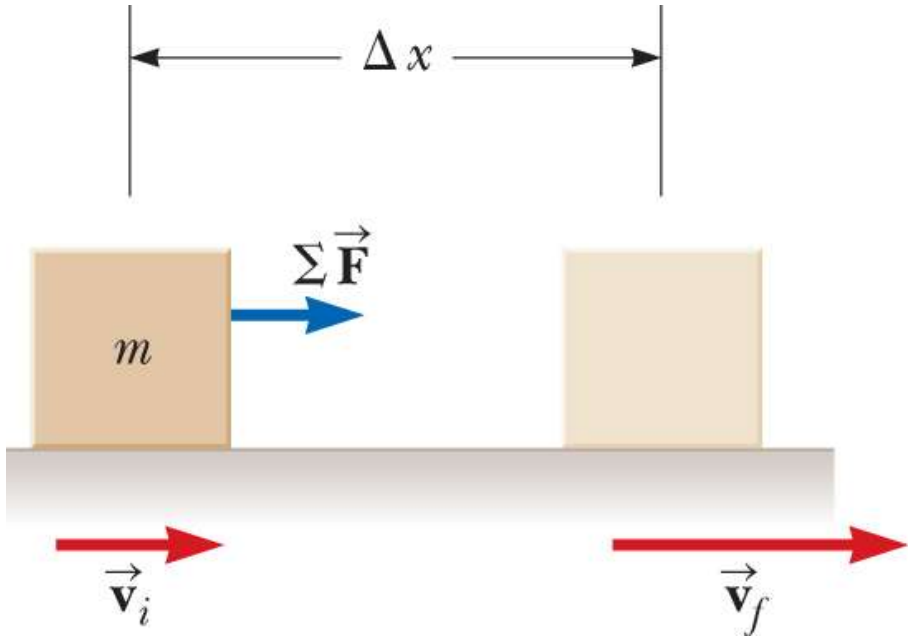
$$W_{\text{ext}} = \int_{x_i}^{x_f} \sum F\,dx$$

$$W_{\text{ext}} = \int_{x_i}^{x_f} ma\,dx = \int_{x_i}^{x_f} m\frac{dv}{dt}dx = \int_{x_i}^{x_f} m\frac{dx}{dt}\frac{dv}{dx}dx = \int_{v_i}^{v_f} mv\,dv$$

$$W_{\text{ext}} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$K \equiv \frac{1}{2}mv^2$$

$$W_{\text{ext}} = K_f - K_i = \Delta K$$



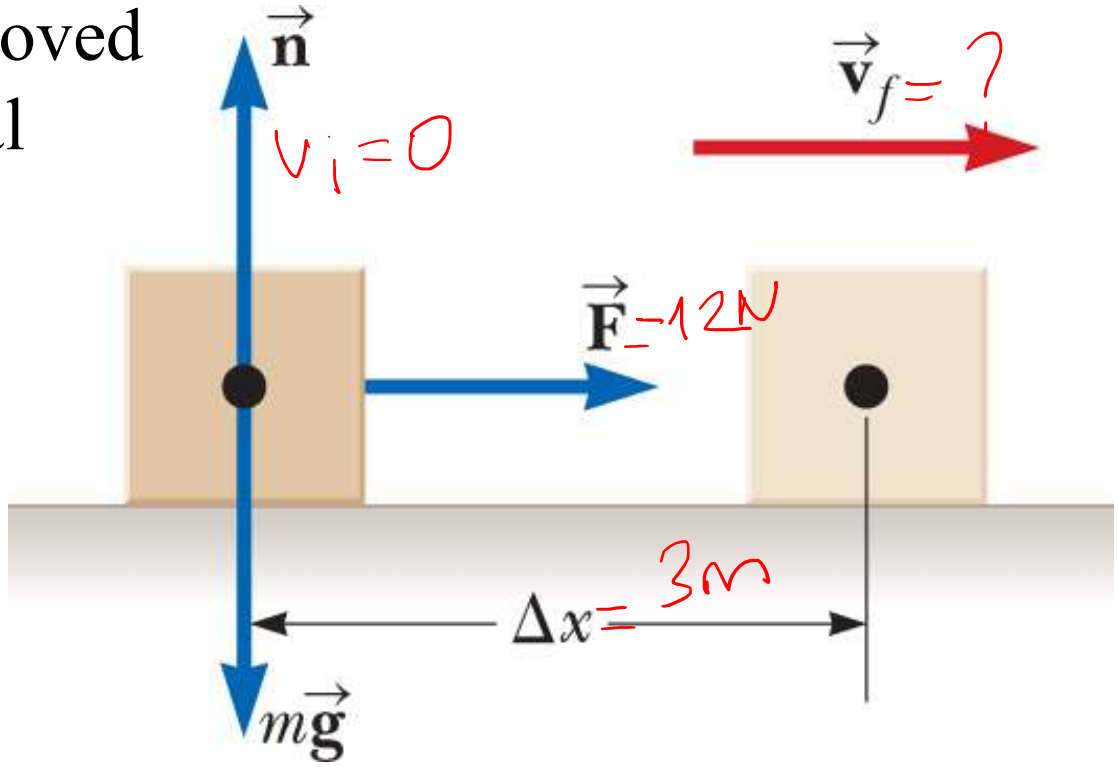
Kinetic Energy and the Work–Kinetic Energy Theorem

$$W_{\text{ext}} = K_f - K_i = \Delta K$$

When work is done on a system and the only change in the system is in the speed of its members, the net work done on the system equals the change in kinetic energy of the system: $W = \Delta K$.

Example 7.6:
A Block Pulled on a Frictionless Surface

A 6.0-kg block initially at rest is pulled to the right along a frictionless, horizontal surface by a constant horizontal force of magnitude 12 N. Find the block's speed after it has moved through a horizontal distance of 3.0 m.



Example 7.6:
A Block Pulled on a Frictionless Surface

$$W_{\text{ext}} = \Delta K = K_f - K_i = \frac{1}{2}mv_f^{\,2} - 0 = \frac{1}{2}mv_f^{\,2}$$

$F\Delta x$

$$v_f = \sqrt{\frac{2W_{\text{ext}}}{m}} = \sqrt{\frac{2F\Delta x}{m}}$$

$$v_f = \sqrt{\frac{2(12\text{ N})(3.0\text{ m})}{6.0\text{ kg}}} = \boxed{3.5\text{ m/s}}$$

Example 7.6:
A Block Pulled on a Frictionless Surface

Suppose the magnitude of the force in this example is doubled to $F' = \Delta F$. The 6.0-kg block accelerates to 3.5 m/s due to this applied force while moving through a displacement $\Delta x'$. How does the displacement $\Delta x'$ compare with the original displacement Δx?

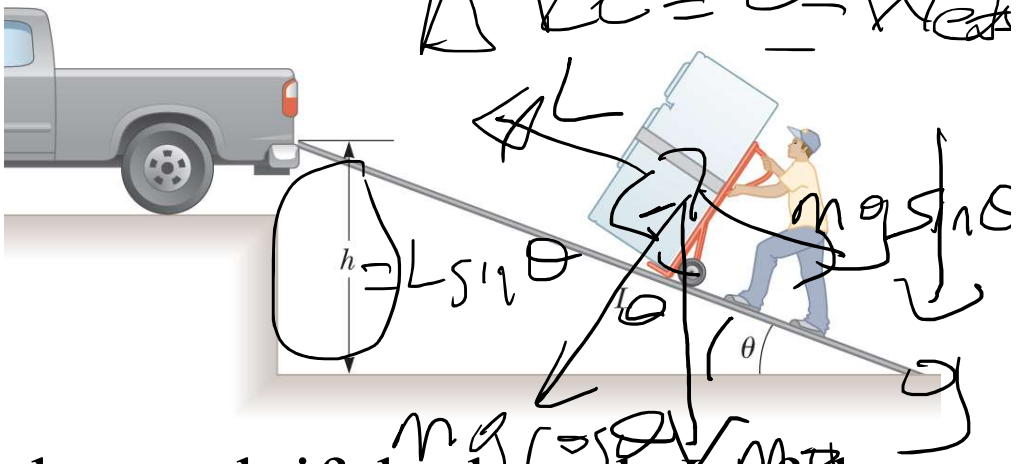
$$\Delta x' < \Delta x$$

$$W_{\text{ext}} = F'\Delta x' = \Delta K = F\Delta x$$

$$\Delta x' = \frac{F}{F'}\Delta x = \frac{F}{2F}\Delta x = \frac{1}{2}\Delta x$$

Conceptual Example 7.7: Does the Ramp Lesson the Work Required?

A man wishes to load a refrigerator onto a truck using a ramp at angle θ as shown in the figure. He claims that less work would be required to load the truck if the length L of the ramp were increased so that the angle θ would be smaller. Is his claim valid? **No.**



$$W_{\text{ext}} = W_{\text{by man}} + W_{\text{by gravity}} = 0$$

$$W_{\text{by man}} = -W_{\text{by gravity}} = -(mg)(L)\left[\cos(\theta + 90^\circ)\right]$$

$$= mgL\sin\theta = mgh$$

Potential Energy of a System

$$\Delta\vec{\mathbf{r}} = \left(y_f - y_i\right)\hat{\mathbf{j}}$$

$$W_{\text{ext}} = \left(\vec{\mathbf{F}}_{\text{app}}\right)\cdot\Delta\vec{\mathbf{r}}$$

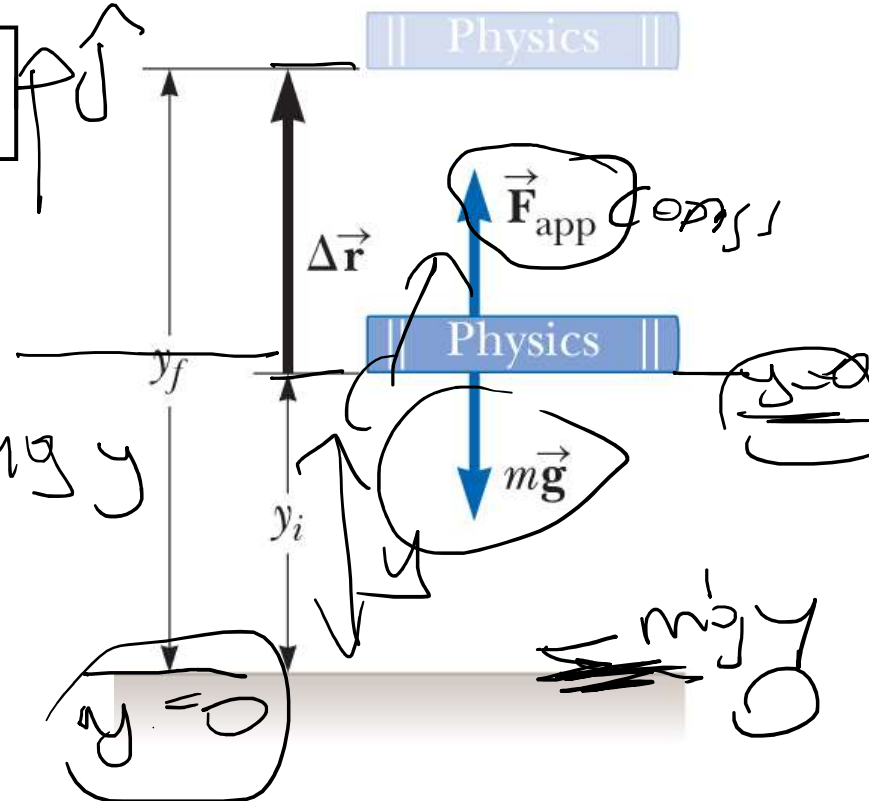
$$= \left(mg\hat{\mathbf{j}}\right)\cdot\left[\left(y_f - y_i\right)\hat{\mathbf{j}}\right]$$

$$= mgy_f - mgy_i$$

$$U_g \equiv mgy$$

$$W_{\text{ext}} = \Delta U_g$$

The work done by the agent on the book–Earth system is $mgy_f - mgy_i$.



Gravitational Potential Energy

Gravitational potential energy: depends only on vertical height of object above surface of Earth



$$W_{\text{ext}} = \left(\vec{\mathbf{F}}_{\text{app}}\right) \cdot \Delta\vec{\mathbf{r}}$$

$$= \left(mg(-\hat{\mathbf{j}})\right) \cdot \left[\left(x_f - x_i\right)\hat{\mathbf{i}} + \left(y_f - y_i\right)\hat{\mathbf{j}}\right]$$

$$= mgy_f - mgy_i$$

$\hat{j} \cdot \hat{i} = 0$

$|\hat{j}||\hat{i}| \cos 90 = 0$

$\hat{j} \cdot \hat{j} = 1$

Quick Quiz 7.6

Choose the correct answer. The gravitational potential energy of a system

(a) is always positive

(b) is always negative

(c) can be negative or positive

$U_G < 0$

> 0

$= 0$

Quick Quiz 7.6

Choose the correct answer. The gravitational potential energy of a system

(a) is always positive

(b) is always negative

(c) can be negative or positive

Example 7.8:
The Proud Athlete and the Sore Toe

A trophy being shown off by a careless athlete slips from the athlete's hands and drops on his foot. Choosing floor level as the $y = 0$ point of your coordinate system, estimate the change in gravitational potential energy of the trophy–Earth system as the trophy falls. Repeat the calculation, using the top of the athlete's head as the origin of coordinates.

Example 7.8:
The Proud Athlete and the Sore Toe

Assume $m_{\text{trophy}} \approx 2$ kg, the top of foot is about 0.05 m above the floor, and trophy falls from a height $h = 1.4$ m.

$$U_i = mgy_i = (2\text{ kg})(9.80\text{ m/s}^2)(1.4\text{ m}) = 27.4\text{ J}$$

$$U_f = mgy_f = (2\text{ kg})(9.80\text{ m/s}^2)(0.05\text{ m}) = 0.98\text{ J}$$

$$\Delta U_g = 0.98\text{ J} - 27.4\text{ J} = -26.4\text{ J} \approx \boxed{-26\text{ J}}$$

$\Delta U = U_f - U_i = -26.$

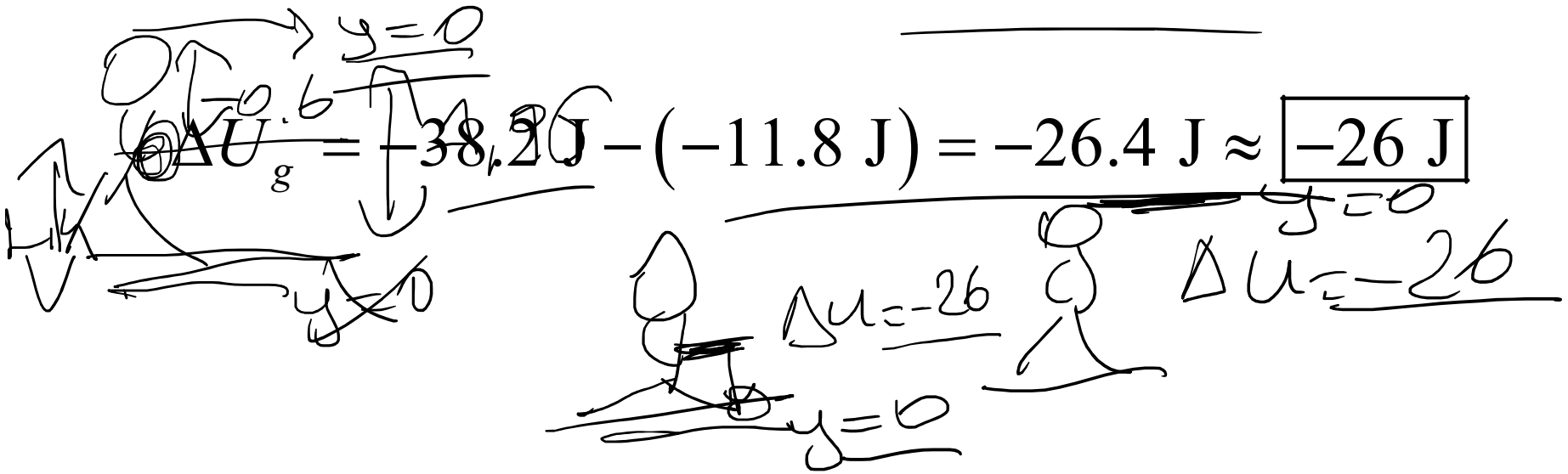
Example 7.8:
The Proud Athlete and the Sore Toe

Assume top of athlete's head is at height $h = 2.0$ m.

$$U_i = mgy_i = (2\text{ kg})(9.80\text{ m/s}^2)(-0.6\text{ m}) = -11.8\text{ J}$$

$$U_f = mgy_f = (2\text{ kg})(9.80\text{ m/s}^2)(-1.95\text{ m}) = -38.2\text{ J}$$

$$\Delta U_g = -38.2\text{ J} - (-11.8\text{ J}) = -26.4\text{ J} \approx \boxed{-26\text{ J}}$$



Elastic Potential Energy

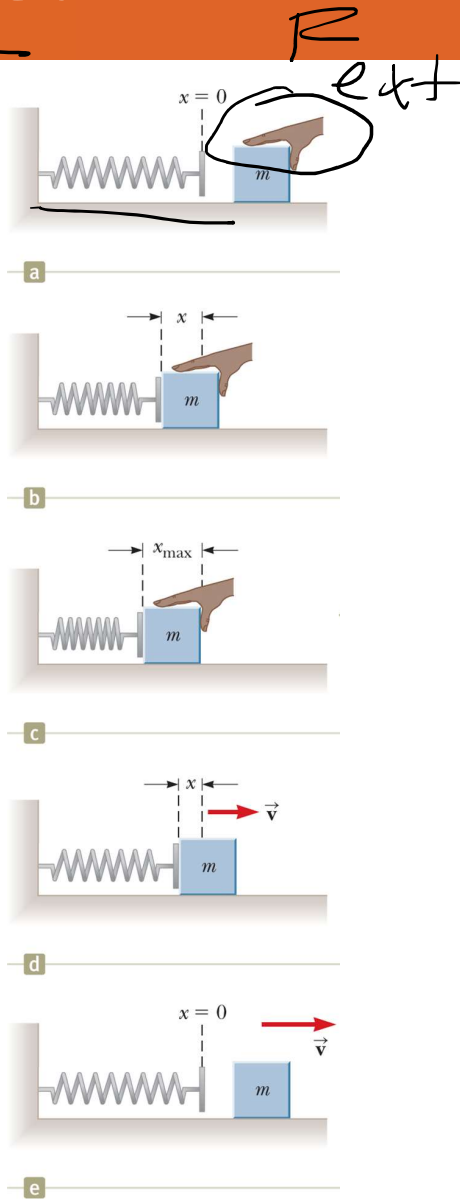
$$W_{\text{ext}} = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2$$

$$U_s \equiv \frac{1}{2}kx^2$$

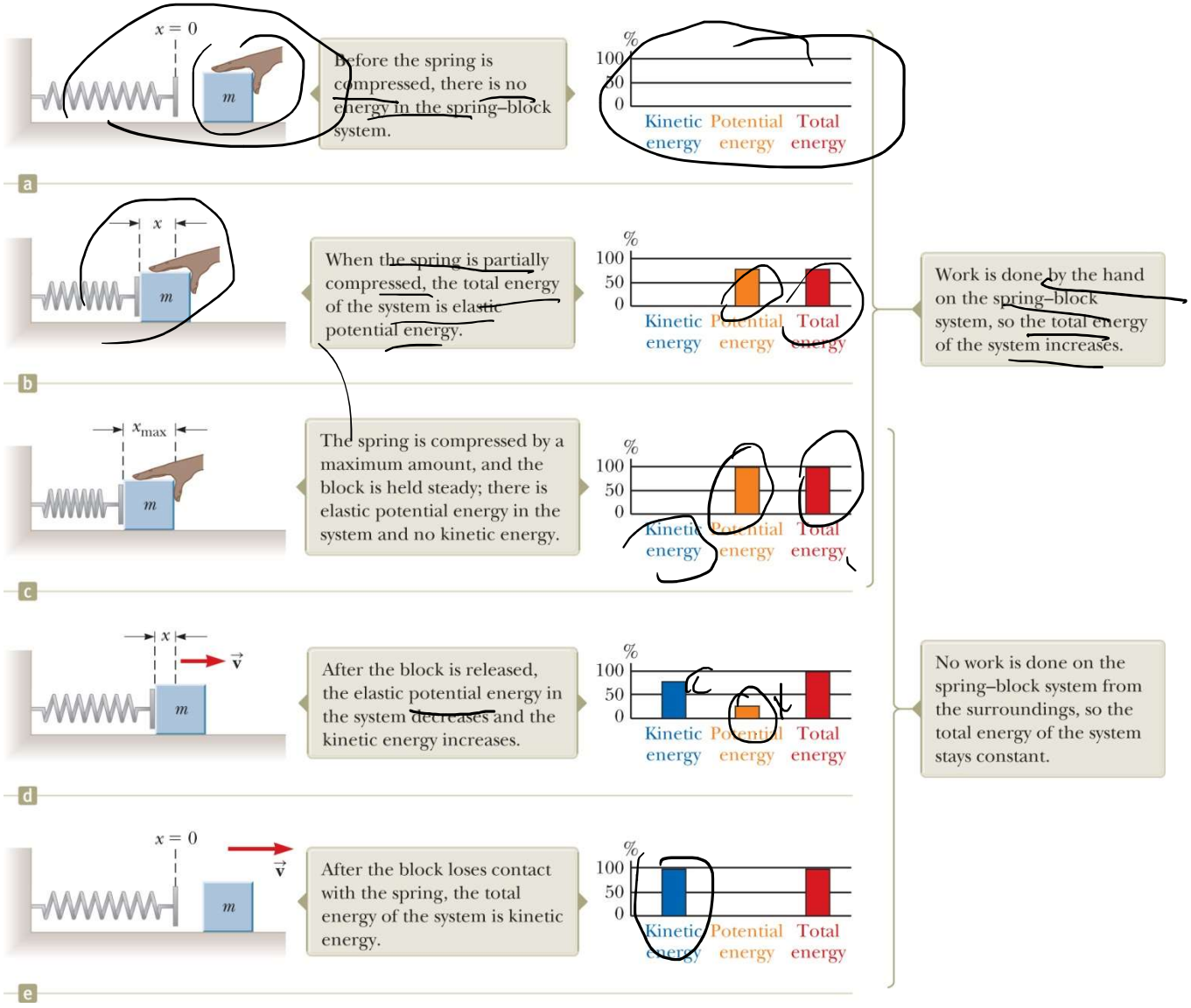
potential energy of spring

$$W_{\text{ext}} = \Delta U_s$$

(external applied hand)

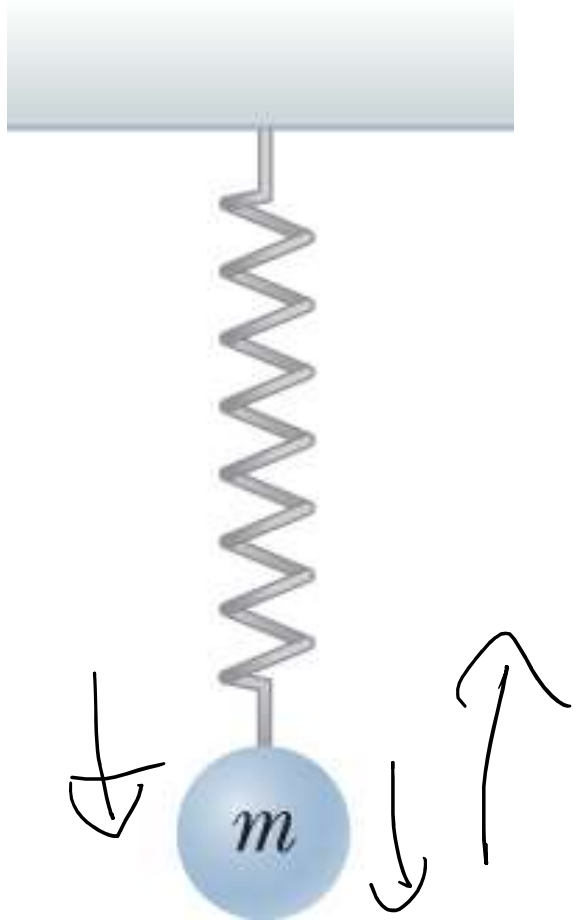


Elastic Potential Energy



Quick Quiz 7.7 Part I

A ball is connected to a light spring suspended vertically as shown in the figure. When pulled downward from its equilibrium position and released, the ball oscillates up and down.



In the system of *the ball, the spring, and the Earth,* what forms of energy are there during the motion?

(a) kinetic and elastic potential

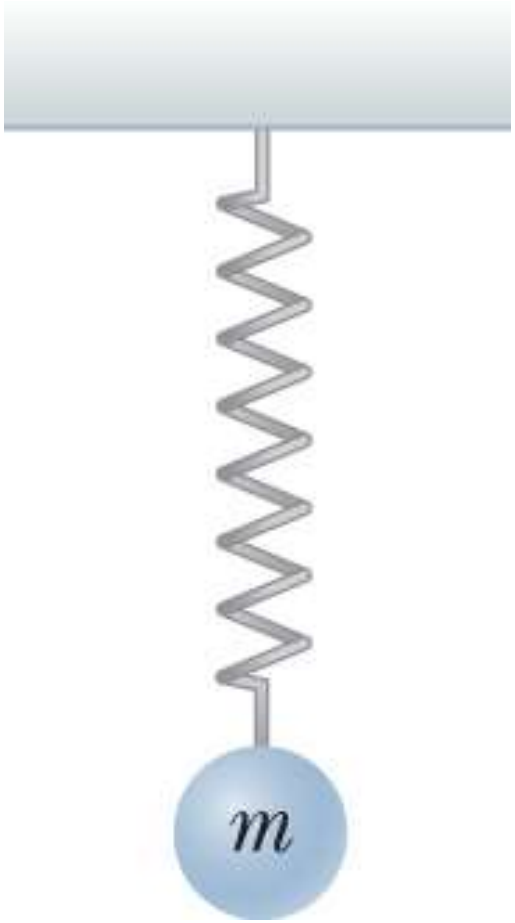
(b) kinetic and gravitational potential

(c) kinetic, elastic potential, and gravitational potential

(d) elastic potential and gravitational potential

Quick Quiz 7.7 Part I

A ball is connected to a light spring suspended vertically as shown in the figure. When pulled downward from its equilibrium position and released, the ball oscillates up and down.



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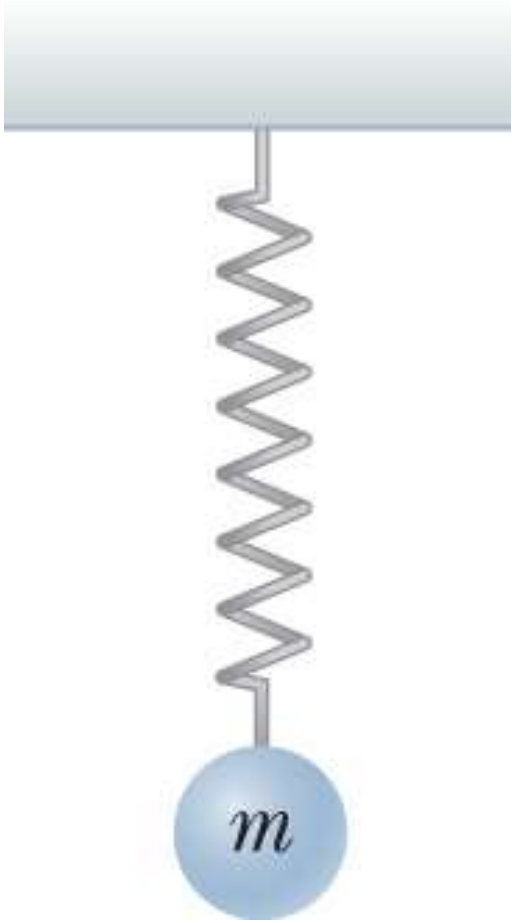
(b) kinetic and gravitational potential

(c) kinetic, elastic potential, and gravitational potential

(d) elastic potential and gravitational potential

Quick Quiz 7.7 Part II

A ball is connected to a light spring suspended vertically as shown in the figure. When pulled downward from its equilibrium position and released, the ball oscillates up and down.



In the system of *the ball and the spring,* what forms of energy are there during the motion?

(a) kinetic and elastic potential

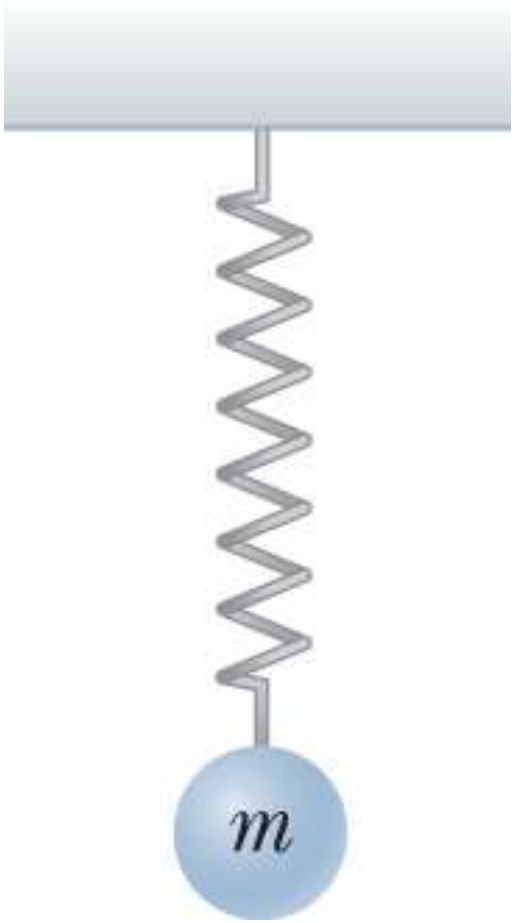
(b) kinetic and gravitational potential

(c) kinetic, elastic potential, and gravitational potential

(d) elastic potential and gravitational potential

Quick Quiz 7.7 Part II

A ball is connected to a light spring suspended vertically as shown in the figure. When pulled downward from its equilibrium position and released, the ball oscillates up and down.



In the system of *the ball and the spring*, what forms of energy are there during the motion?

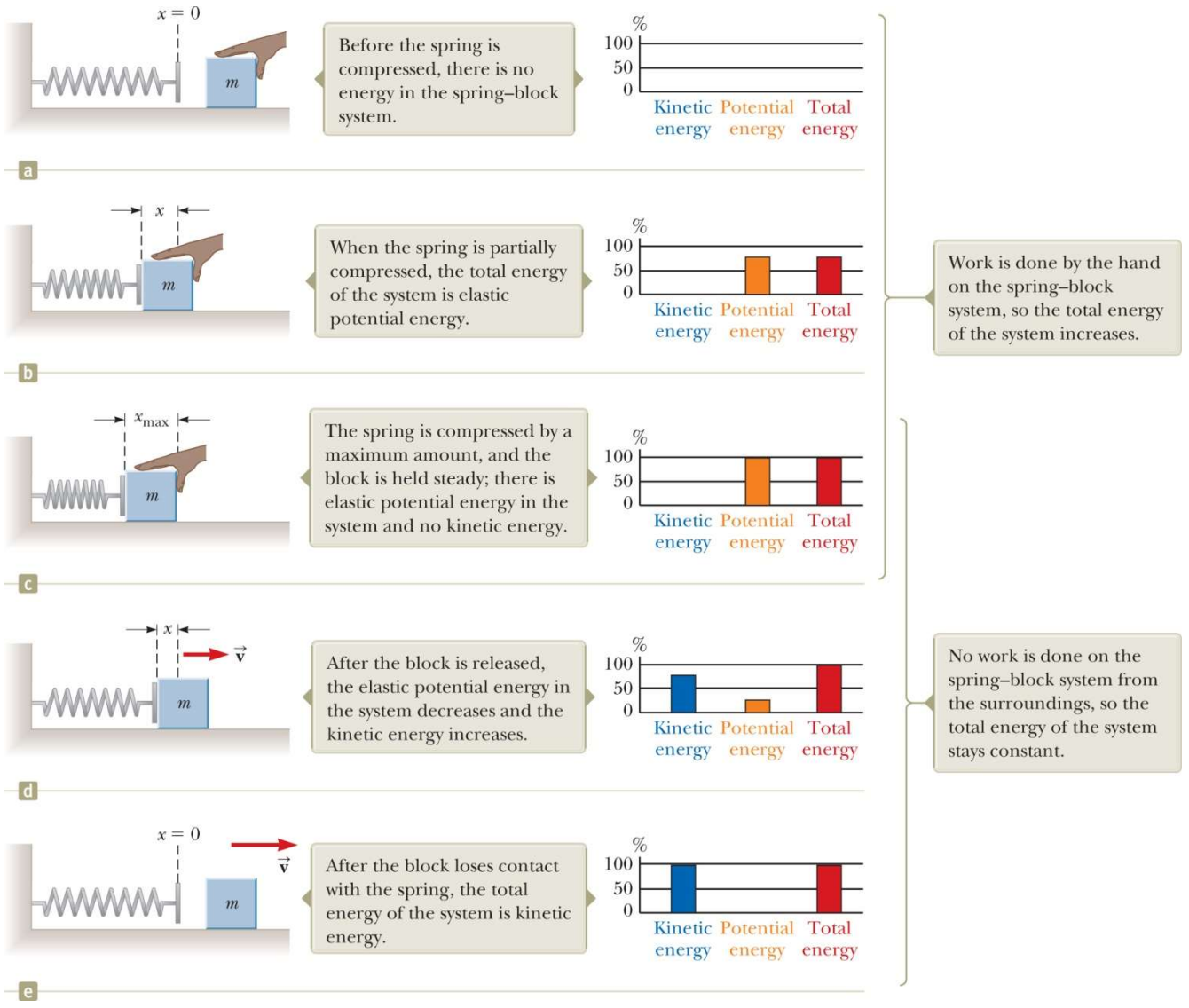
(a) kinetic and elastic potential

(b) kinetic and gravitational potential

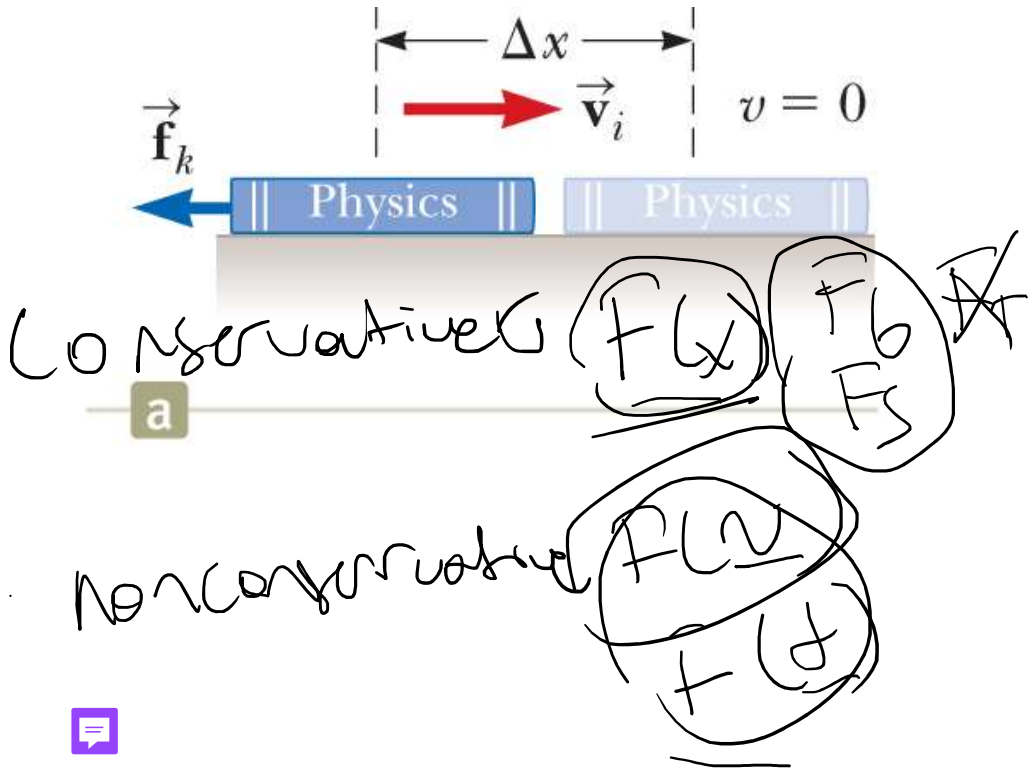
(c) kinetic, elastic potential, and gravitational potential

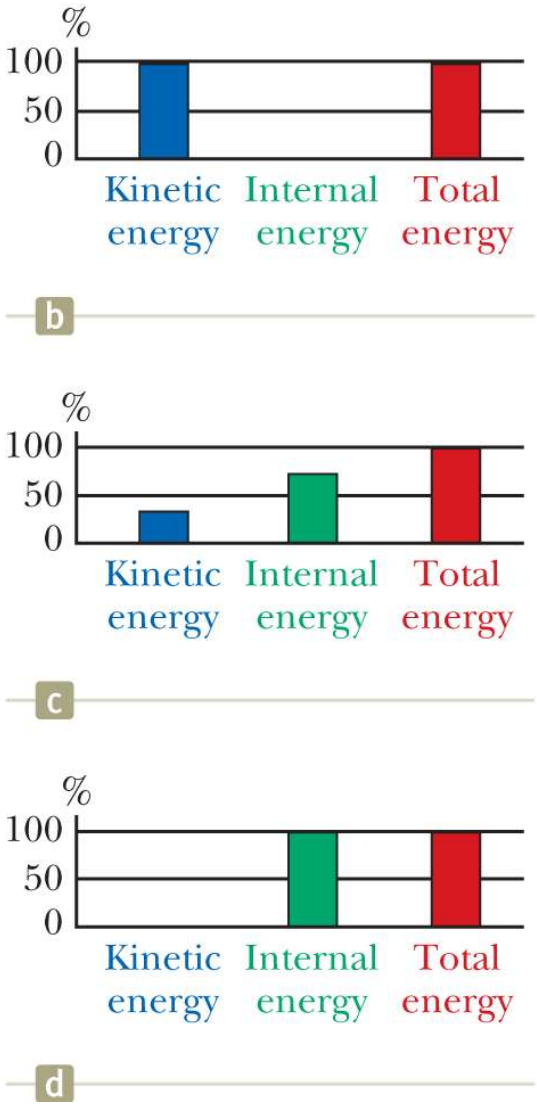
(d) elastic potential and gravitational potential

Energy Bar Charts

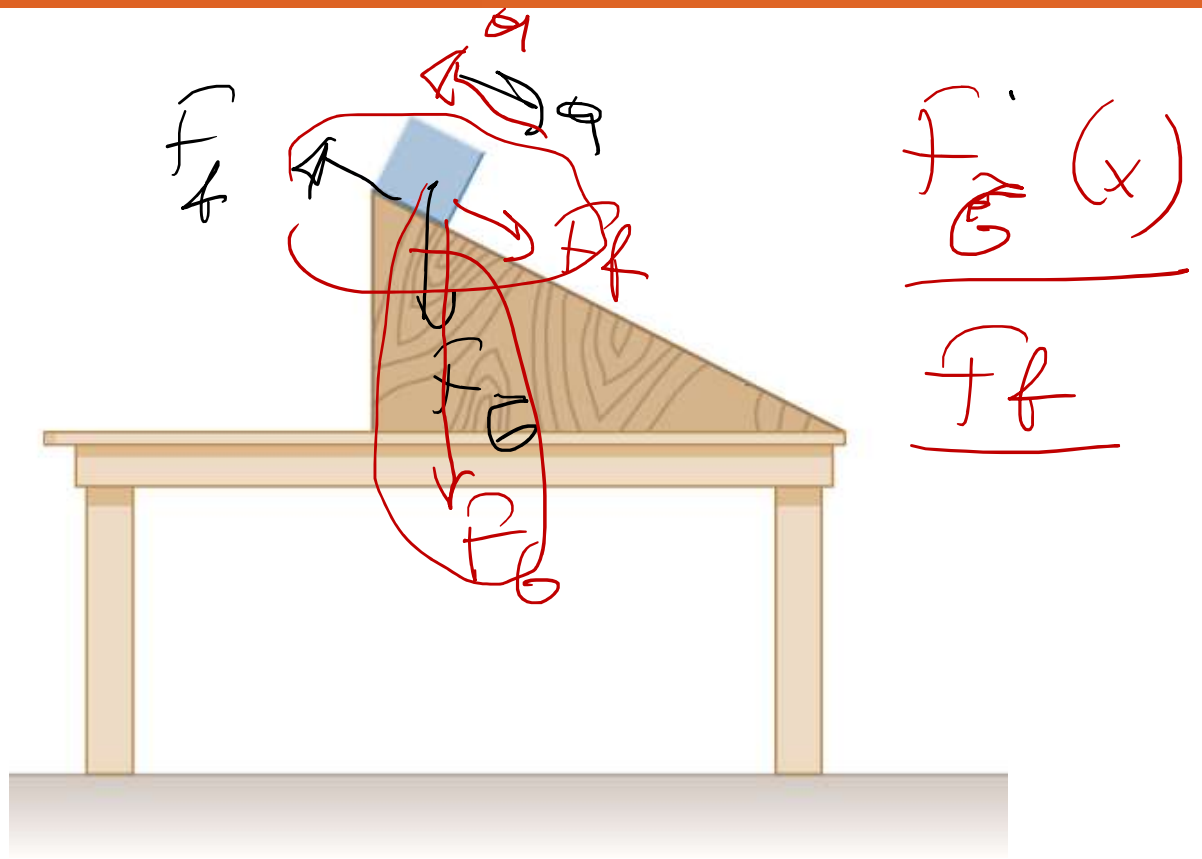


Conservative and Nonconservative Forces





Conservative and Nonconservative Forces



Gravitational force: conservative

Friction force: nonconservative

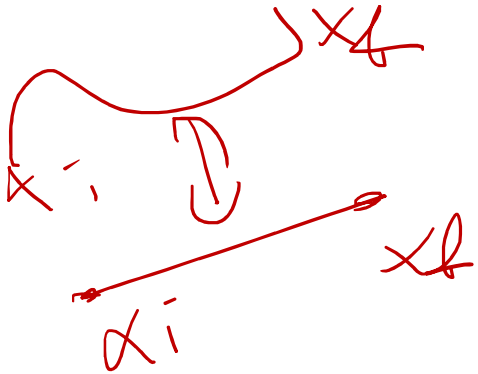
Conservative Forces

1. Work done by a conservative force on a particle moving between any two points is independent of the path taken by the particle.

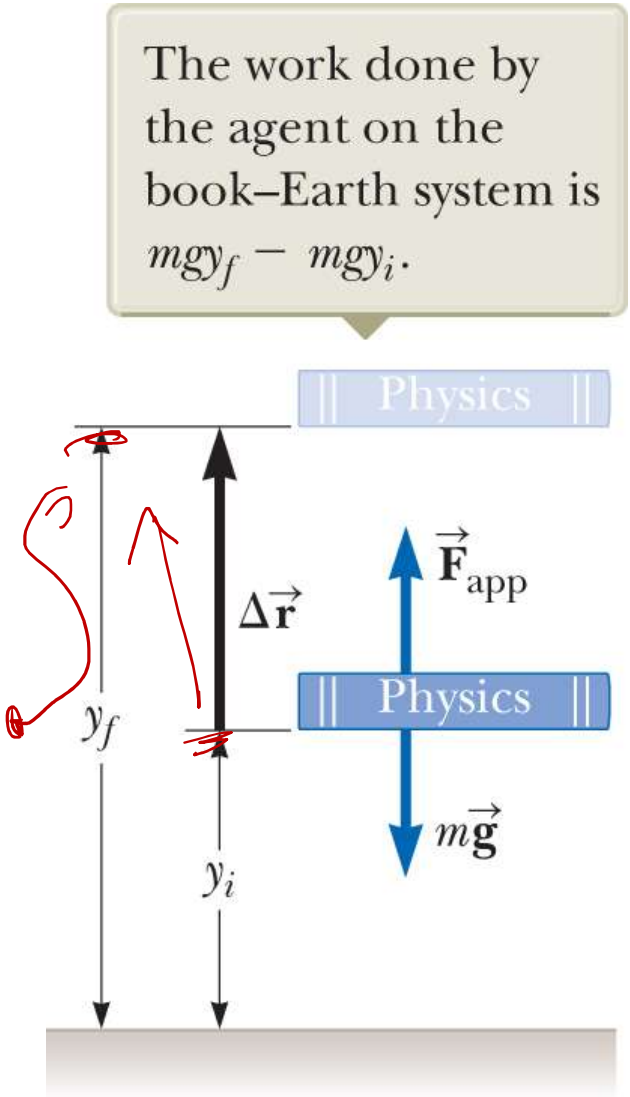
2. Work done by a conservative force on a particle moving through any closed path is zero.

$$W_g = -mg\hat{\mathbf{j}} \cdot \left[\left(y_f - y_i \right) \hat{\mathbf{j}} \right] = mgy_i - mgy_f$$

$$W_s = \frac{1}{2} k x_i^{\,2} - \frac{1}{2} k x_f^{\,2}$$



Nonconservative Forces

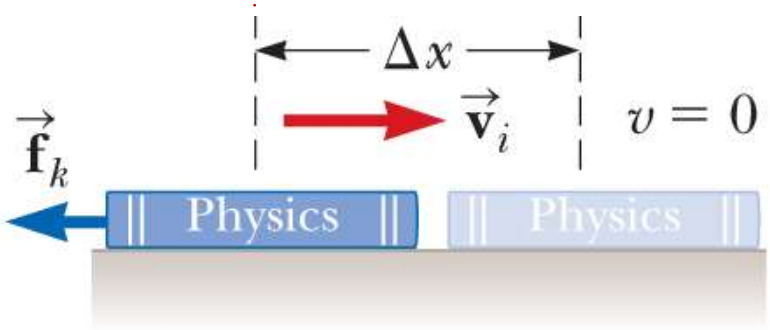


$K+U=$ Mechanical energy

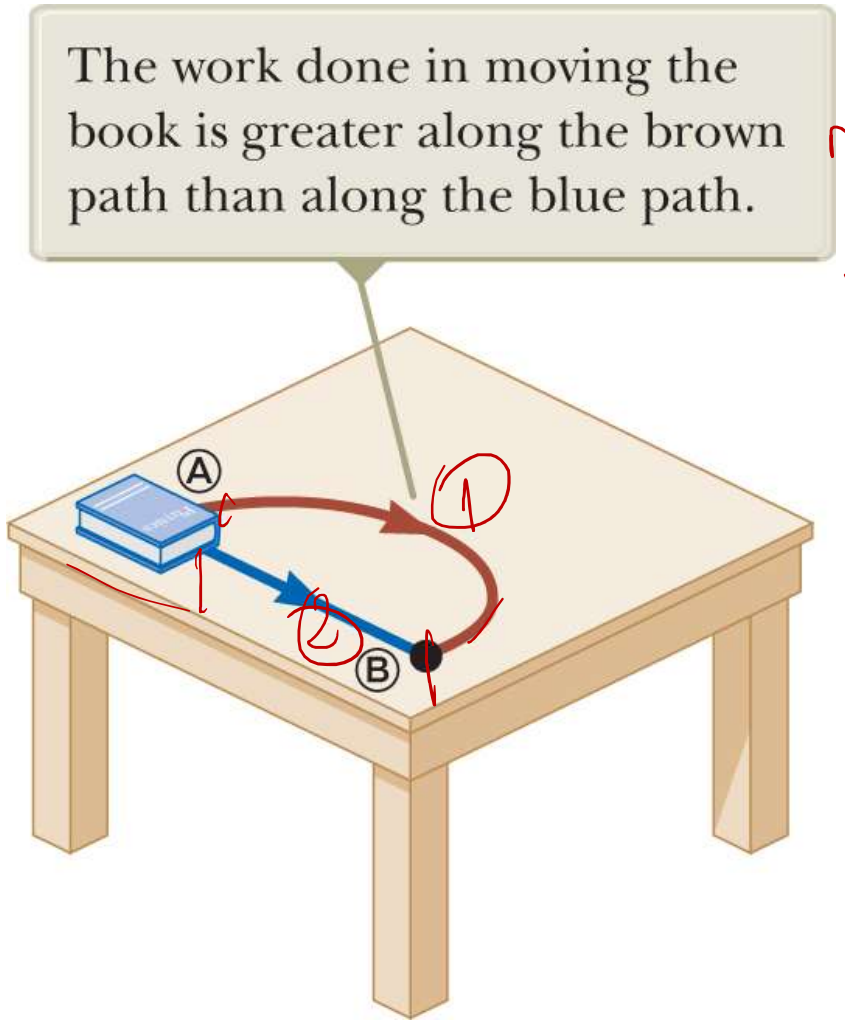
$$E_{mech} \equiv K + U$$

mechanical

Work done by friction?



Nonconservative Forces



nonconservative force

$W_1 \neq W_2$

conservative

$W_1 = W_2$

Relationship Between Conservative Forces and Potential Energy

$$W_{\text{int}} = U_i - U_f = -\Delta U$$

$$W_{\text{int}} = \int_{x_i}^{x_f} F_x \, dx = -\Delta U$$

$$\Delta U = U_f - U_i = -\int_{x_i}^{x_f} F_x \, dx$$

$$U_f(x) = \int_{x_i}^{x_f} F_x \, dx + U_i$$

$$dU = -F_x \, dx$$

$$F_x = -\frac{dU}{dx}$$

Quick Quiz 7.8

What does the slope of a graph of $U(x)$ versus x represent?



(a) The magnitude of the force on the object

(b) the negative of the magnitude of the force on the object

(c) the x component of the force on the object

(d) the negative of the x component of the force on the object

Quick Quiz 7.8

What does the slope of a graph of $U(x)$ versus x represent?

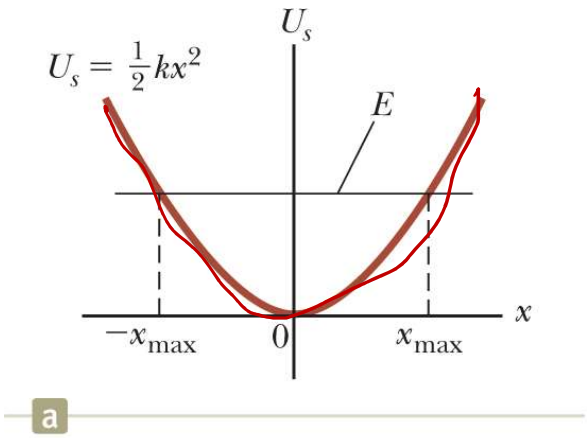
(a) The magnitude of the force on the object

(b) the negative of the magnitude of the force on the object

(c) the x component of the force on the object

(d) the negative of the x component of the force on the object

Energy Diagrams and Equilibrium of a System



$x = 0$ position for a block–spring system:
stable equilibrium

